

Weather Station with Data Logging Project

Biomedical Engineering - Microprocessors Project

1. Project Overview

This project is a Weather Station with Data Logging built using the PIC16F877A microcontroller. The system measures temperature, humidity, light intensity, date, and time. The collected data is displayed on an LCD screen and transmitted to a computer through UART for logging and monitoring.

2. Components Used

- PIC16F877A Microcontroller
- DHT11 Temperature and Humidity Sensor
- LDR (Light Dependent Resistor)
- DS1307 Real Time Clock (RTC)
- 16x2 LCD Display
- Red and Green LEDs
- Push Button
- UART Serial Communication

2. System Operation

The DHT11 sensor measures temperature and humidity. The LDR measures ambient light intensity through the ADC module. The DS1307 RTC provides the current date and time. All measurements are displayed on the LCD and transmitted through UART for data logging.

3. Data Logging

Data logging is achieved using UART communication at 9600 baud rate. The sensor readings are sent to a PC and viewed using serial terminal software such as PuTTY. The logged data includes temperature, humidity, light intensity, date, and time.

4. User Interface

A push button is used to switch between two LCD screens. The first screen displays weather information, while the second screen displays the current date and time.

5. LED Indication

The red LED turns ON when the temperature exceeds 30°C, indicating a high-temperature condition. The green LED turns ON when the temperature is within the normal range.

6. Interfaces Used

- ADC Interface: Reads the LDR value.
- I2C Interface: Communicates with the DS1307 RTC.
- UART/USART Interface: Sends data to the PC.
- Single-Wire Communication: Communicates with the DHT11 sensor

7. Conclusion

This project demonstrates sensor interfacing, real-time monitoring, data logging, ADC conversion, UART communication, and I2C communication using the PIC16F877A microcontroller. It provides a complete embedded weather monitoring system suitable for educational and laboratory applications.